The documentation process conversion measures necessary to comply with this revision shall be completed by 29 August 1997

METRIC

MIL-PRF-19500/610C
29 May 1997
SUPERSEDING
MIL-S-19500/610B
22 July 1994

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, HERMETIC, DIODE, SILICON, SCHOTTKY BARRIER,
TYPES 1N6677-1 AND 1N6677UR-1
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, Schottky barrier diodes. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance for die (element evaluation).

1.2 Physical dimensions. See figure 1 (DO-35), figure 2 (DO-213AA), and figure 3 (JANC die) dimensions.

1.3 Maximum ratings.

Types	V_{RWM}	I_{O1} 1/	I_{FSM}	T_{STG}	T_J
	V (pk)	mA dc	A (pk)	C°	C°
1N6677-1, 1N6677UR-1	40	200	5	-65 to +150	-65 to +125

1/ Derate linearly at 8 mA/°C above T_L or T_{EC} = +100°C where T_L is at L = 9.52 mm (.375 inch).

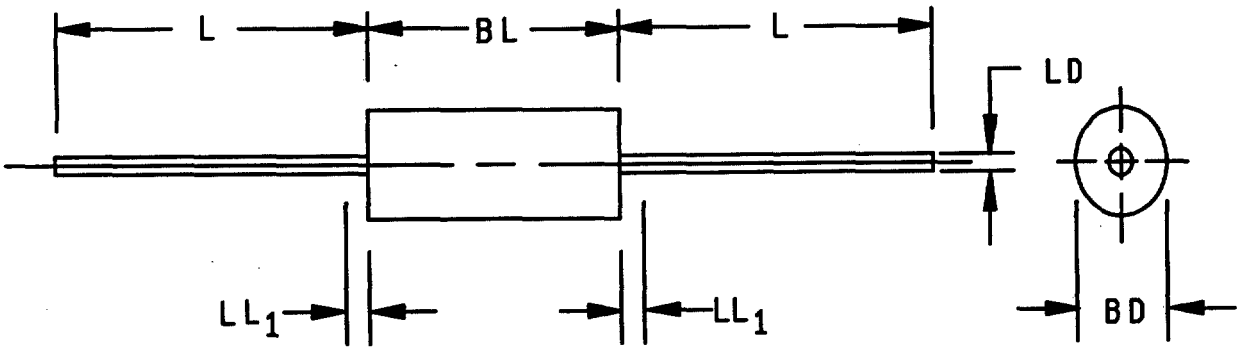
1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics at T_A = +25°C.

Types	Max V_{FM1}	Max V_{FM2}	Max V_{FM3}	Max I_{RM} V_{RM} = 40 V dc pulsed method (see 4.5.1)		Max $R_{\theta JL}$ or $R_{\theta JEC}$ 9.52 mm (.375") lead length or end cap	Max $Z_{\theta JX}$	Max C_T
	I_{FM} = 20 mA	I_{FM} = 200 mA	I_{FM} = 630 mA	T_J = +25°C I_{RM1}	T_J = +100°C I_{RM2}			V_R = 0 V dc
	V (pk)	V (pk)	V (pk)	µA	mA	°C/W	°C/W	pF
1N6677-1	0.37	0.50	0.70	5.0	0.60	250	20	50
1N6677UR-1	0.37	0.50	0.70	5.0	0.60	100	20	50

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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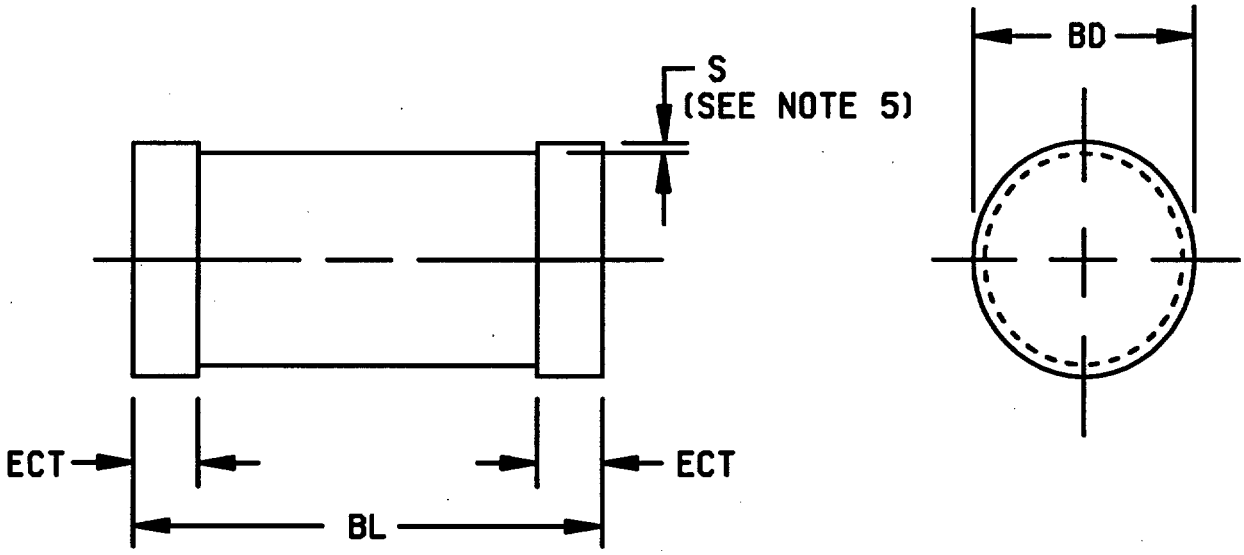


Symbol	Dimensions				Notes
	Millimeters		Inches		
	Min	Max	Min	Max	
LD	0.46	0.56	.018	.022	
BD	1.40	2.29	.055	.090	3
BL	3.05	5.08	.120	.200	3
L	25.40	38.10	1.000	1.500	
LL_1		1.27		.050	4

NOTES:
1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. Package contour optional within BD and length L. Heat slugs, if any, shall be included within this cylinder but shall not be subject to minimum limit of BD.
4. Within this zone lead, diameter may vary to allow for lead finishes and irregularities other than heat slugs.

FIGURE 1. Physical dimensions of 1N6677-1 (DO-35).

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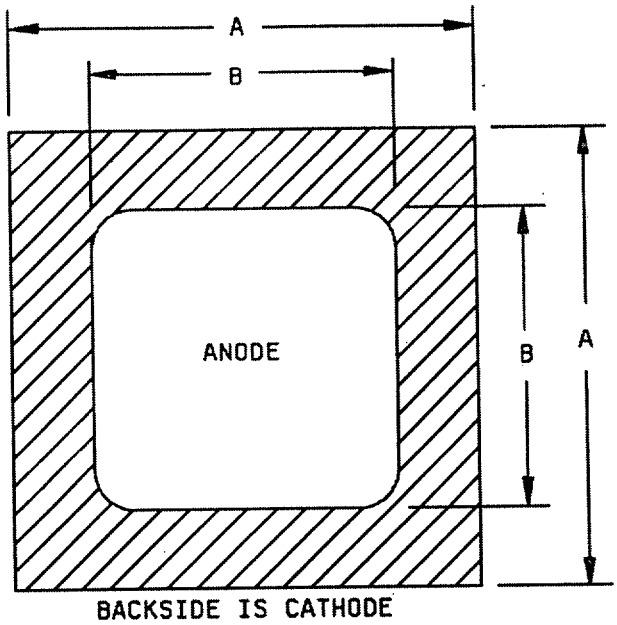


Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
BD	1.60	1.70	.063	.067
ECT	0.41	0.56	.016	.022
BL	3.30	3.71	.130	.146
BG_1	2.54 ref		.100 ref	
S	0.03 min		.001 min	

NOTES:
1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.

FIGURE 2. Physical dimensions of surface mount family, 1N6677UR-1 (DO-213AA).

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A version

Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
A	0.53	0.64	.021	.025
B	0.33	0.43	.013	.017

Design data

Metallization:
Top: (Anode) . . . . . AL
Back: (Cathode). . . . . Au

AL thickness . . . . . 25,000 Å minimum
Gold thickness . . . . . 4,000 Å minimum
Chip thickness . . . . . 0.254 mm (10 mils) ± 0.051 mm (2 mils)

FIGURE 3. JANC (A-version) die dimensions.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified on figures 1, 2, and 3 herein. Plastic packages are prohibited. The US Government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-PRF-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.5).

3.3.2 Diode construction. All devices shall be metallurgically bonded, double plug construction in accordance with the requirements of category I or II (see MIL-PRF-19500).

3.3.2.1 Surface mount. The U version shall be considered structurally identical to the non-U version devices except for lead attach.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

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3.4.1 Marking for UR devices. UR-suffix parts are to be marked with a cathode band only. Initial container package marking will be in accordance with MIL-PRF-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 JANHC and JANKC devices. Qualification for JANC devices shall be in accordance with appendix G of MIL-PRF-19500. This testing may be performed on a TO-5 package in lieu of the axial leaded package.

4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurement		
	JANS level	JANTX and JANTXV levels	JAN level
3a	Temperature cycling	Temperature cycling	Temperature cycling in accordance with MIL-PRF-19500, JANTX level.
3c 1/	Thermal impedance (see 4.5.3)	Thermal impedance (see 4.5.3)	Thermal impedance (see 4.5.3)
9	I_{R1} and V_{FM2}	Not applicable	Not applicable
10 2/	T_A = +120°C; V_{RWM} = 40 V(pk); I_O = 0, half sine wave, f = 60 Hz	T_A = +120°C; V_{RWM} = 40 V(pk); I_O = 0, half sine wave, f = 60 Hz	Not applicable
11	ΔI_{R1} ± 100 percent of initial reading or 2 µA, whichever is greater. $\Delta V_{FM2} \leq \pm$ 50 mV dc.	I_{R1} and V_{FM2}	Not applicable
12	See 4.3.2	See 4.3.2, t = 48 hours	Not applicable
13	Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or 2 µA, whichever is greater; $V_{FM2} \leq \pm$50 mV dc	Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or 2 µA, whichever is greater; $V_{FM2} \leq \pm$50 mV dc.	Not applicable

1/ Thermal impedance shall be performed any time after sealing provided. Temperature cycling is performed in accordance with appendix E, table IV, screen 3 of MIL-PRF-19500 prior to this thermal impedance.

2/ Junction temperature (T_J) is not to exceed +125°C with V_{RWM} = 40 V(pk). T_J is affected by the device mounting thermal resistance when parasitic power is generated by the temperature dependent leakage current. Until this leakage becomes significant near thermal runaway, T_J remains approximately equal to T_A or T_L for I_O = 0.

4.3.1 Screening (JANHC or JANKC). Screening of die shall be in accordance with appendix G, of MIL-PRF-19500. As a minimum, die shall be 100 percent probed in accordance with group A, subgroup 2.

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4.3.2 Burn-in conditions. Burn-in conditions are as follows: I_F = 200 mA dc. Mounting and test conditions in accordance with MIL-STD-750, method 1038, test condition B.

4.3.2.1 Mounting. Devices may be mounted using any convenient method including the temporary attachment of leads on US suffix devices, provided that the parts are burned-in at $T_J \geq +110°C$.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500 and table I herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a. I_M measurement current ..... 1 mA to 10 mA.

b. I_H forward heating current ..... 0.5 A to 1.0 A.

c. t_H heating time ..... 10 ms.

d. t_{MD} measurement delay time ..... 70 µs maximum.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIa (JANS), and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
3	4066	I_{FSM} = 5 A(pk), condition A 2, I_O = 200 mA; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); 5 surges of 8.3 ms each at 1 minute intervals.
4	1036	I_O = 200 mA; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); f = 50-60 Hz; V_{RWM} = 40 V(pk); $t_{on} = t_{off}$ 3 minutes minimum for 2,000 cycles.
5	1027	$T_A = +100°C \pm 25°C$; I_O = 200 mA (minimum) with I_O adjusted as required to achieve an average lot $T_J = +125°C$ +0°C, -5°C.
6	3101	$R_{\theta JL}$ = 250°C/W; L = 9.52 mm (.375 inch) lead length (non-surface mount). $R_{\theta JEC}$ = 100°C/W (surface mount).

4.4.2.2 Group B inspection, appendix E, table VIb (JAN, JANTX, and JANTXV of MIL-PRF-19500).

Subgroup	Method	Condition
2	4066	I_{FSM} = 5 A(pk), condition A 2, I_O = 200 mA; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); 5 surges of 8.3 ms each at 1 minute intervals.
3	1027	I_O = 200 mA; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); f = 50-60 Hz; V_{RWM} = 40 V(pk).
4	2075	In accordance with 4.5.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

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4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
2	2036	Tension: Test condition A; weight = 8 pounds; t = 15 s. Lead fatigue: Test condition E; weight 1 pounds. NOTE: Both tension and lead fatigue are not applicable for US devices.
6	1026	I_O = 200 mA; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); f = 50-60 Hz; V_{RWM} = 40 V(pk).

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with appendix E, table IX of MIL-PRF-19500 and the conditions for subgroup testing in table II herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in 4.3.2.1 of MIL-STD-750.

4.5.2 Steady-state operation life. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 150°.

4.5.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit (not to exceed the group A, subgroup 2 limit) for $Z_{\theta JX}$ in screening (appendix E, table V of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition.

4.5.3.1 For initial qualification or requalification. Read and record data ($Z_{\theta JX}$) shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum). Twenty-two serialized devices shall be sent to the qualifying activity for test correlation.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.5.3	$Z_{\theta JX}$		20	°C/W
Forward voltage	4011	I_{FM} = 20 mA (pk) pulse method (see 4.5.1)	V_{FM1}		0.37	V
	4011	I_{FM} = 200 mA (pk) pulse method (see 4.5.1)	V_{FM2}		0.50	V
	4011	I_{FM} = 630 mA (pk) pulse method (see 4.5.1)	V_{FM3}		0.70	V
Reverse current leakage	4016	V_{RM} = 40 V (pk) pulse method (see 4.5.1)	I_{RM1}		5.0	µA
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	V_{RM} = 40 V (pk) pulse method (see 4.5.1)	I_{RM2}		0.60	mA
Forward voltage	4011	I_F = 200 mA (pk) pulse method (see 4.5.1)	V_{FM4}		0.47	V
Low temperature operation:		T_A = -55°C				
Reverse current leakage	4016	V_{RM} = 40 V (pk) pulse method (see 4.5.1)	I_{RM3}		20	µA
Forward voltage	4011	I_F = 200 mA (pk) pulse method (see 4.5.1)	V_{FM5}		0.60	V
Subgroup 4						
Capacitance	4001	V_R = 0 V dc .01 ≤ f ≤ 1 MHz V_{SIG} = 15 mV p-p	C_T		50	pF

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group E inspection (all quality levels) for qualification only. 1/

Inspection 2/	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Thermal shock (temperature cycling)	1051	500 cycles	
Hermetic seal	1071	Test condition E	
Electrical measurement		See table I, group A, subgroup 2	
Subgroup 2			22 devices c = 0
Steady-state reverse bias	1038	test condition A; 1,000 hours, See 4.3, screen 10	
Electrical measurement		See table I, group A, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance, (forward voltage drop diode method)	4081 or 3101	$R_{\theta JL}$ = 250°C/W maximum at 9.52 mm (.375 inch) lead length; $R_{\theta JEC}$ = 100°C/W maximum; method 3101 in accordance with 4.5.3 except I_H = 200 mA and T_H = 20 s (minimum)	c = 0
Subgroup 5			
Not applicable			
Subgroup 6			
Not applicable			

1/ For initial design and process change verification only (one time testing).
2/ For sampling plan, see MIL-PRF-19500.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N6677) will be identified on the QPL.

JANC ordering information		
	Manufacturer	
PIN	55801	---
1N6677-1	JANHCA1N6677-1	---
	JANKCA1N6677-1	---

6.5 Ordering data. Acquisition documents may specify the material and finish.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-1902-06)

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/610C	2. DOCUMENT DATE (YYMMDD) 970529

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, HERMETIC, DIODE, SILICON, SCHOTTKY BARRIER, TYPES 1N6677-1 AND 1N6677UR-1 JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

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c. ADDRESS (Include Zip Code) Defense Electronics Supply Center Attn: DESC-ELDT Dayton, Ohio 45444-5765	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** *Defense Quality and Standardization Office* 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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